Alexander M. Lapuente

Email: lapuente@bu.edu
Phone: (704)-654-5819
LinkedIn: alex-lapuente

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Education

Boston University — Boston, MA, USA
Ph.D. in Physics, GPA: 3.65/4.00. Advisor: Kirit S. Karkare — September 2024–Present

- Coursework: Cosmology; Observational Astronomy; Particle Physics; QFT I-II

University of Chicago — Chicago, IL, USA
B.A. w/ Honors in Physics, B.S. in Math., GPA: 3.79/4.00. Advisor: David W. Miller — September 2020–June 2024

- Thesis: *Daily Modulation Search for Uniformly Polarized Dark Photon Dark Matter*
- Math Coursework: Abstract Algebra I-II; Real Analysis I-III; Complex Analysis; Topology
- Physics Coursework (graduate level): Adv. Math Methods; Cosmology; Particle Phys.; QFT I-III

United States Particle Accelerator School — Melville, NY, USA
USPAS Summer School — June 2023

- Coursework: Fundamentals of Accelerator Physics and Technology

Research

I am interested in exploring the universe's large-scale structure and evolutionary history via line intensity mapping (LIM). This research program targets atomic and molecular emission lines from unresolved galaxies at (sub-)millimeter wavelengths, promising unique access to the high-redshift universe. I am interested in the detector technologies used to conduct these observations as well as the analysis of data products. I am also interested in simulation work to model LIM signals and study instrumental systematics.

SuperSpec — May 2025–Present

- Assisted in Summer 2025 SuperSpec-MUSCAT (SuMAC) deployment at Large Millimeter Telescope. Performed hardware testing, calibration measurements, and preliminary analysis.
- Characterization of detector noise performance; flux calibration; atmospheric modelling; data reduction and analysis pipeline design.

SPT-SLIM — January 2025–Present
South Pole Telescope Shirokoff Line Intensity Mapping

- Created low-level data visualization and analysis modules for detector performance diagnostics.
- Creation and analysis of combined sky-maps.
- Implemented atmospheric calibration method.
- Performed principal component analysis (PCA) to study noise sources and correlation.

LIM Simulations — January 2025–Present

- Use Bayesian methods for mm-wave galaxy signal estimation. Use Markov Chain Monte Carlo and Fisher information methods to simulate signal detection and reconstruction. Sensitivity study to determine how quality of detector profile measurements impacts parameter estimation.
- Use `pysides` module to simulate galactic dust emission spectra and radiative transfer.

Past Research

GigaBREAD — Department of Physics, University of Chicago
Gigahertz-range Broadband Reflector Experiment for Axion Detection — March 2022–June 2024

- Phase I: Dark Photon Search. Developed and tested signal amplification chain and DAQ system. Conducted noise temperature, S-parameter, etc. measurements. Tested and improved RFI masking. Wrote DAQ scripts in `Python` using custom RFSoC firmware. Modularized DAQ and analysis routines.
- Phase II: Axion Search. Interfaced piezoelectric motor to DAQ system. Tested amplifier performance within external $\vec{B}$-field. Improved RFI exclusion regime in noisier environment. Prepared for initial data run.

PIP-II — Beamline Instrumentation Group, FermiLab
Proton Improvement Plan II — May 2023–August 2023

- Developed versatile method of simulating proton-beam signals. Simulated transmission of induced current through resistive wall current monitor. Signal processing in `Matlab` and `Python` to study beam properties. Determined accuracy bounds for time-domain signal reconstruction.
- Developed `Python` program to provide live data from DC current transformer in booster ring. Containerized project using Docker and integrated it with existing Redis database.

Publications

[1] C. S. Benson et al., "Spectral characterization and performance of SPT-SLIM on-chip filterbank spectrometers", IEEE Transactions on Applied Superconductivity, 1–7 (2026) 10.1109/TASC.2026.3694583.

[2] K. R. Dibert et al., "An On-Sky Atmospheric Calibration of SPT-SLIM", (2025).

[3] G. Hoshino et al. (GigaBREAD Collaboration), "First axionlike particle results from a broadband search for wavelike dark matter in the 44 to 52μeV range with a coaxial dish antenna", Phys. Rev. Lett. **134**, 171002 (2025) 10.1103/PhysRevLett.134.171002.

[4] M. R. Young et al., *Design and Performance of the SPT-SLIM Receiver Cryostat*, 2025.

[5] S. Knirck et al. (BREAD Collaboration), "First results from a broadband search for dark photon dark matter in the 44–52 μeV range with a coaxial dish antenna", Phys. Rev. Lett. **132**, 131004 (2024) 10.1103/PhysRevLett.132.131004.

[6] E. E. Piano and C. E. Piano, "Bargaining over beauty: the economics of contracts in renaissance art markets", Journal of Law and Economics **66** (2023).

Presentations, Talks, and Posters

[1] A. M. Lapuente et al., *Data Reduction Pipeline for the SuMAC Millimeter-Wave Spectrometer at the LMT*, Poster presented at SPIE Astronomical Telescopes+Instrumentation 2026, July 8, 2026.

[2] A. M. Lapuente, *Spectral Calibration Requirement for Line Recovery*, Talk presented at SPT-SLIM annual collaboration meeting, June 10, 2024.

[3] A. M. Lapuente, *GigaBREAD at Argonne National Laboratory*, Talk presented at BREAD annual collaboration meeting, Oct. 4, 2023.

[4] A. M. Lapuente, *Signal Simulation and Analysis for PIP-II RWCM and DCCT Systems*, Poster presented at FermiLab Summer intern poster session, Aug. 4, 2023.

[5] A. M. Lapuente, *GigaBREAD Data Acquisition System*, Talk presented at BREAD annual collaboration meeting, Aug. 30, 2022.

Featured Media

[1] L. Lerner, *First Results from BREAD Experiment Demonstrate a New Approach to Searching for Dark Matter*, Apr. 2, 2024.

Teaching

Boston University — Boston, MA, USA
Teaching Fellow (Introductory Mechanics) — January 2026–May 2026

The University of Chicago — Chicago, IL, USA
Teaching Assistant (Calculus I-II-III) — September 2021–June 2024

Skills

- **Operating Systems:** Windows, Ubuntu
- **Software:** Docker/Podman, Redis
- **Hardware:** FPGA, MKID, RFSoc
- **Physics:** Experimental design; data collection, analysis, & representation; RF-engineering and signal processing

Languages

- **Natively Spoken:** English
- **Fluently Spoken:** Spanish, Latin (Classical)
- **Fluently Typed:** LaTeX, Mathematica, Python
- **Poorly Typed:** HTML, MatLab

Honors and Awards

- Boston University Photonics Center Student Travel Support Grant — 2026
- Boston University Dean's Fellowship (Resarch Grant) — 2024–2025
- University of Chicago Dean's List — 2024
- Jeff Metcalf Scholarship (Summer Research Grant) — 2022, 2023
- University of Chicago Fermi Scholarship (Summer Research Grant) — 2020–2024
- University of Chicago President's Merit Scholarship (Summer Research Grant) — 2020–2024
- University of Chicago University Merit Scholarship (Study Grant) — 2020–2024